

Chapter 1

Questions from February 12, 2025

Question 1

We have defined

$$\psi_x^{(m)} = \delta_{xy} = \begin{cases} 1, & \text{if } x = y \\ 0, & \text{else} \end{cases}.$$

It is stated in [Str06, page 35]¹:

“Since f is assumed continuous and the support of $\psi_x^{(m)}$ is close to x then,

$$\int_K f \psi_x^{(m)} d\mu \approx f(x) \int_K \psi_x^{(m)} d\mu \quad (1.1)$$

.”

But $\text{supp}(\psi_x^{(m)}) = \{x\}$, in which case i think it would be

$$\int_K f \psi_x^{(m)} d\mu = \int_x f d\mu = 0$$

but this is not the case, so why does (1.1) hold? And further don't we still need to integrate f ?

This question is mostly asked to figure out why $\frac{\int_K f \psi_x^{(m)} d\mu}{\int_K \psi_x^{(m)} d\mu}$ converges uniformly to f .

Question 2

When showing that

$$\sum_{x \in V_m \setminus V_0} v(x) f_m(x) \psi_x^{(m)} \quad (1.2)$$

converges uniformly to $v(x)f(x)$ [Str06, eq. (2.2.10)], i have run into a few problems.

1. What is the value of (1.2), because $\forall \omega \in V_m \setminus V_0$ i would assume (when showing the uniform convergence) that

$$\left| \left(\sum_{x \in V_m \setminus V_0} v(x) f_m(x) \psi_x^{(m)}(\omega) \right) - v(\omega) f(\omega) \right| = |v(\omega) f_m(\omega) - v(\omega) f(\omega)|.$$

so i am mostly just asking if i have understood the sum correctly.

2. Assuming that the above understanding of the sum is correct, i have gotten

$$\begin{aligned} \left| \left(\sum_{x \in V_m \setminus V_0} v(x) f_m(x) \psi_x^{(m)}(\omega) \right) - v(\omega) f(\omega) \right| &= |v(\omega) f_m(\omega) - v(\omega) f(\omega)| \\ &= |v(\omega)| |f_m(\omega) - f(\omega)| \end{aligned}$$

¹Robert S. Strichartz. *Differential Equations on Fractals: A Tutorial*. Princeton University Press, 2006

to finish this will use the fact that $f_m(\omega) \rightarrow f(\omega)$ uniformly to get some $\frac{\epsilon'}{|v(\omega)|}$ to use as bound. But the this bound uses ω as variable, so i would like some upper bound of $|v(\omega)|$ to use and i can not quite figure out if there is a bound. This is probably due to my poor understanding of what i means that $v \in \text{dom}_0 \mathcal{E}$.

3. Lastly in [Str06, eq (2.2.9)] he moves the summation into the integral, which i am assuming is due do the $\mathcal{L}^1(\mu)$ measurability of $v(\omega), f_m(\omega)$